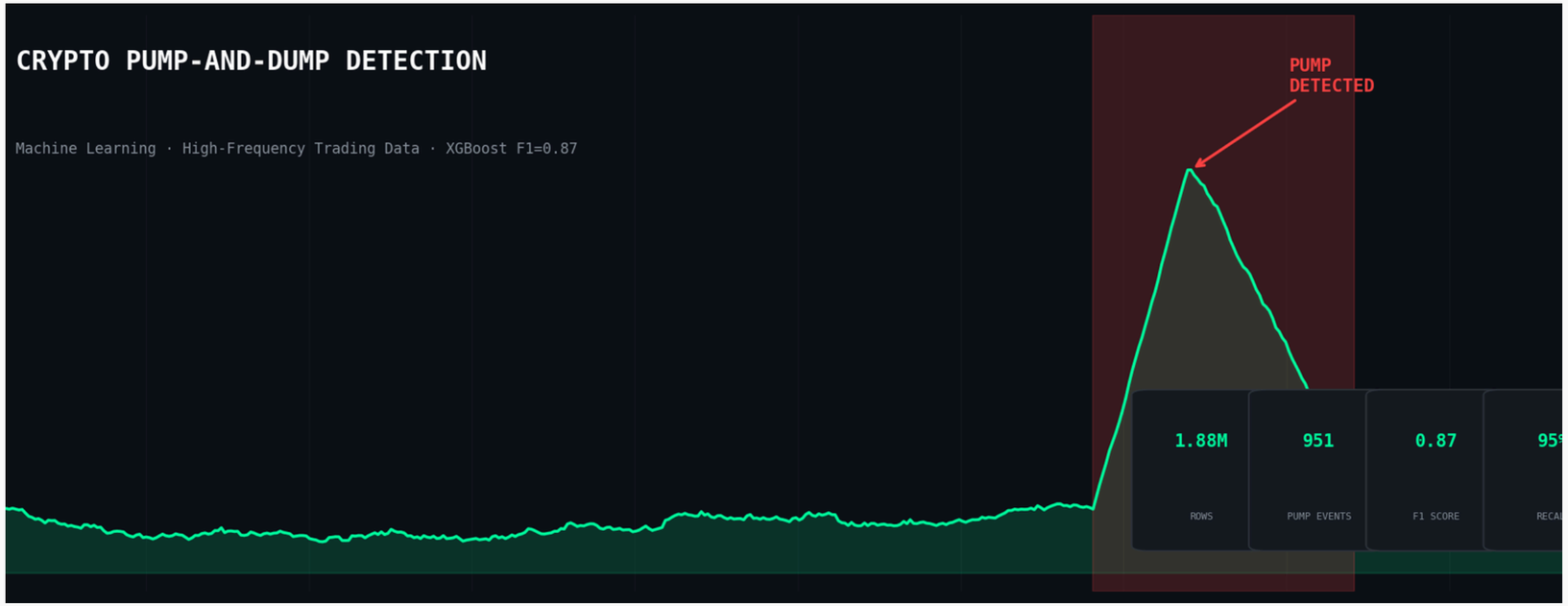
Crypto Pump-and-Dump Detection

Group 2

Isabella S
Nicole O.
Alvin
S.Kimani



Problem Statement

Problem Statement

The Problem

- Crypto markets are vulnerable to coordinated pump-and-dump schemes
- Artificial price inflation followed by rapid sell-offs
- Leads to financial losses and market distortion
- Detection is challenging due to extreme class imbalance

Project Objective:

Objective

Primary Goal:

- Develop a machine learning model to detect pump events using short time-window trading features

Secondary Goals:

- Handle extreme class imbalance
- Compare multiple models
- Use time-based validation
- Evaluate using precision-recall metrics

.

Dataset Overview

Dataset Description.

Raw trades were grouped into 5s, 15s and 25s to reduce noise and capture short time behaviour patterns. Each row represents a summary of trading activity in a short time period for a specific cryptocurrency.

High-frequency crypto market data at:

- 5-second intervals
- 15-second intervals
- 25-second intervals

Target Variable:

- gt = 1 → Pump event
- gt = 0 → Normal activity

Pump events represent only 0.0657% of the data.

Why Multiple Time Windows?

Why Multiple Time Windows?

Why 5s, 15s, 25s?

- 5s →Very sensitive to rapid coordinated buying meaning it detects micro-spikes (early signal)
- 15s → Captures short bursts more reliably
- 25s → Detects sustained abnormal activity, it’s a bit delayed but more stable.

Insight:

5s detects early spikes → 15s confirms escalation → 25s confirms trend.

Key Predictive Features

Key Feature Categories

Volume & Activity

- std_volume captures volume explosion
- avg_volume captures volume explosion average
- std_trades captures spike in trading activity

Price Dynamics (captures rapid acceleration)

- std_price
- avg_price
- avg_price_max

Order Intensity (captures sudden buying pressure)

- std_rush_order
- avg_rush_order

Time

- hour(cos and sin) Minute(cos and sin) to preserve cyclical time patterns.

Exploratory Data Analysis

All three data sets were combined for Data analysis and modelling.

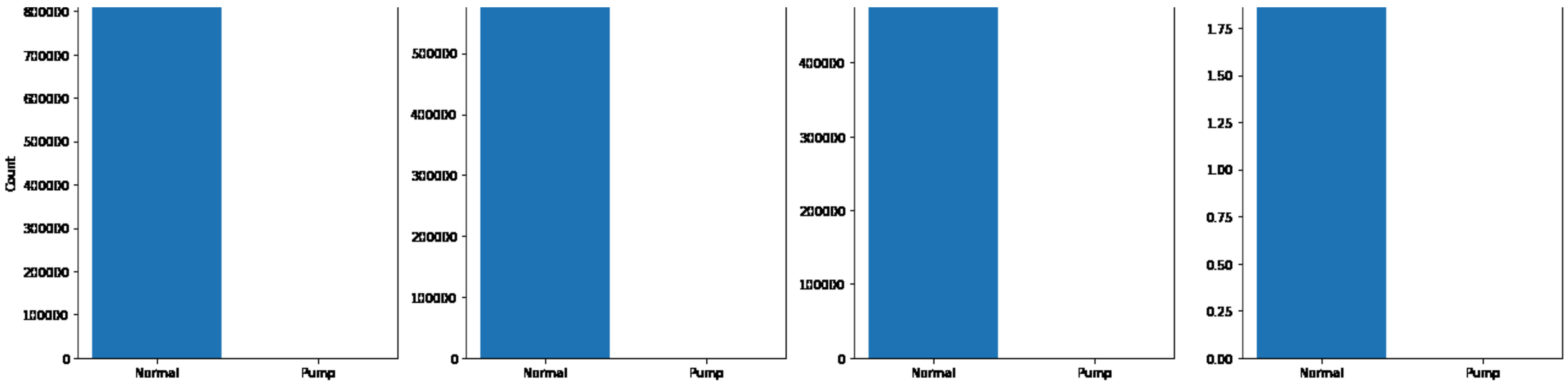
Pump vs Normal Behavior

Pump events show:

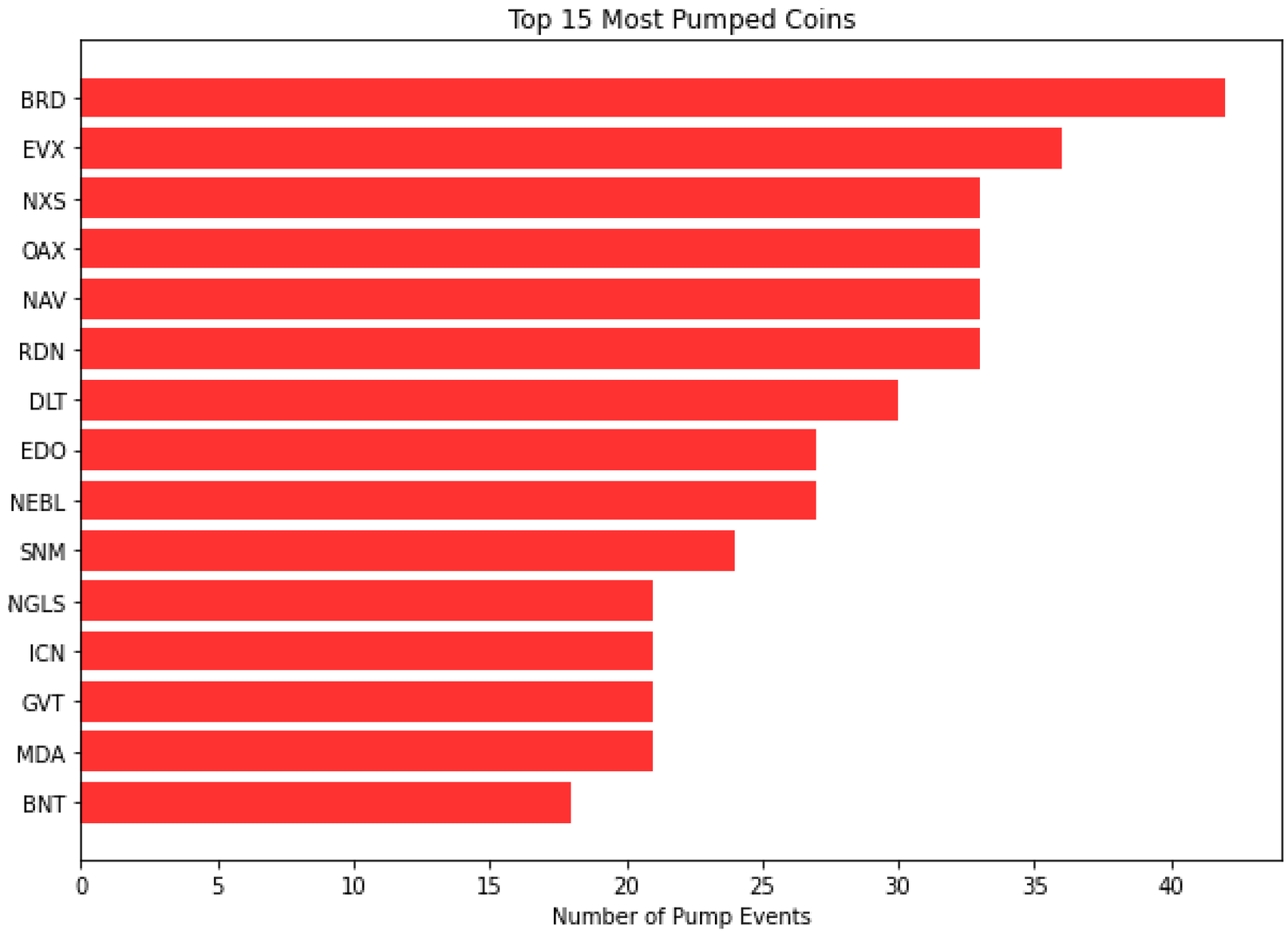
- Much higher std_rush_order
- Significant volume spikes
- Increased trade variability

Example:

std_rush_order during pump ≈ 50–100× higher than normal



This shows class imbalance across all three datasets



BRD – Bread (BRD Token),EVX – Everex, NXS – Nexus, QAX – Quantstamp (QAX Token), NAV – NavCoin, RDN – Raiden Network Token, DLT – Agrello (DLT Token) are the top most pumped coins.

Time-Based Validation

Why Not Random Split?

- Data is time-series
- Random split causes temporal leakage
- Model could “learn from the future”

Solution:

- Train on data before May 2020
- Test on data after May 2020

Modelling

Models Evaluated

1. Logistic Regression
2. Random Forest
3. XGBoost

Evaluation Focus:

- Precision (Class 1)
- Recall (Class 1)
- F1-Score
- Precision-Recall tradeoff

Results.

Model	Precision	Recall	F1-Score
Logistic Reg.	0.12	1	0.22
Random Forest	0.71	0.95	0.82
XGBoost	0.79	0.95	0.87

Conclusions

- Among the models evaluated, XGBoost demonstrated the strongest performance in identifying suspicious pump activity while minimizing false alarms. Its ability to handle class imbalance and capture nonlinear relationships made it the most suitable model for practical monitoring applications.
- BRD – Bread (BRD Token),EVX – Everex, NXS – Nexus, QAX – Quantstamp (QAX Token), NAV – NavCoin, RDN – Raiden Network Token, DLT – Agrello (DLT Token) are the top most pumped coins. These assets may be more vulnerable to coordinated price manipulation.
- Because pump events are rare compared to non pump events, traditional accuracy measures can be misleading. Instead, performance was evaluated based on the model's ability to correctly identify suspicious events without generating excessive false alarms.
- Volume and order intensity are strong pump indicators

Recommendations

- Cryptocurrency exchanges should consider implementing automated monitoring systems that leverage machine learning models to detect abnormal short-term price and volume spikes. Early detection of suspicious activity can help reduce market manipulation, and protect platform integrity.
- Investors should exercise caution when trading low-liquidity tokens that exhibit sudden and extreme price increases. Such patterns may indicate coordinated pump activity, increasing the risk of financial loss.
- continuous model monitoring and periodic retraining are recommended to maintain predictive performance. Machine learning-based monitoring could strengthen enforcement capabilities and improve transparency within digital asset markets.

Thank You For Your Time